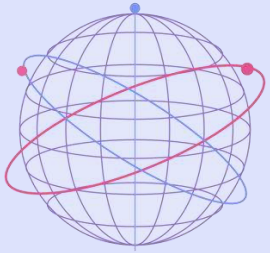
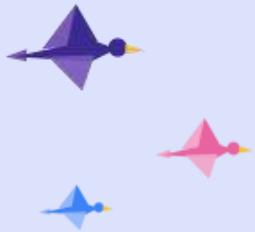
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
Iron Lab: Quantum-Cryptographic Agent Communications

TRACK
Quantum Cryptograpy And Security

TEAM NAME
IRON LAB

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2.Somappa Nandi
3.Somalingappa M kumbar

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COUNTRY
INDIA

Problem Statement

In 2025, AI agents started communicating via GibberLink acoustic protocol for 10x efficiency, switching from human language to FSK sound beeps. This is now used in autonomous systems, but current GibberLink transmission is in plain text - any microphone can eavesdrop and decode sensitive data like financial orders and tactical communications.

At the same time, quantum cryptography faces critical real-world challenges:

Quantum Eavesdropping: Standard QKD protocols like BB84 (Prepare & Measure) and E91 (Entanglement) are vulnerable to hardware attacks. Photon Number Splitting (PNS) attack allows Eve to steal key information while keeping QBER low. Detector Blinding forces Bob's detector to click on Eve's command.

Active Tampering: Passive eavesdropping is not the only threat. Active Eve can alter the message in transit - Man-in-the-Middle tampering. Without integrity checks, corrupted packets are accepted.
Channel Loss & Fragility: Real quantum channels face 60%+ attenuation due to distance, clouds in satellite QKD, and fiber loss. Pure QKD fails and aborts, leaving no secure fallback.
No PQC + QKD Hybrid with AI Context: NIST released Kyber-768 ML-KEM (FIPS 203) in Aug 2024 for post-quantum security, but no platform combines it with QKD for AI-to-AI communication with live tamper defense

Objectives & Novelty

OBJECTIVES:
Build Iron Lab Hybrid Protocol (QKD BB84/E91 + NIST Kyber-768 PQC)
Secure AI-to-AI GibberLink acoustic communication via AES-256-GCM
Simulate Eve with Passive (PNS) & Active Tampering attacks (1-100%)
Live tamper detection & packet drop on auth tag mismatch
Test under variable qubits & channel loss conditions

NOVELTY:
First quantum-secured GibberLink for AI agents
Hybrid auto-fallback controller - never fails insecure
Interactive Attack Marketplace - judges can launch attacks live
Live Integrity Alert - shows TAMPERING DETECTED & Eve Failed to Forge Auth Tag as in dashboard
4-layer enterprise dashboard - protocol + transmission + Eve + live security

Solution & Quantum Advantage

SOLUTION:

Iron Lab is a hybrid quantum-safe platform where Agent Alice (A) and Agent Bob (B) negotiate a shared key via Iron Lab Hybrid Protocol - QKD (BB84/E91 via Qiskit 2.6 Aer) + NIST Kyber-768 PQC. Final key = SHA3(PQC XOR QKD). Message is encrypted with AES-256-GCM, converted to GibberLink FSK acoustic beeps, transmitted. Bob verifies GCM auth tag. If Eve enables Active Tampering (e.g., PNS Mu attack), Bob's dashboard instantly flags SECURITY_EXCEPTION: AES-256-GCM AUTHENTICATION TAG MISMATCH and drops packet, showing EVE FAILED TO FORGE VALID AUTH TAG. Trust controller monitors QBER & loss, auto-switches to PQC_ONLY if QBER >11% or loss >60%.

QUANTUM ADVANTAGE:

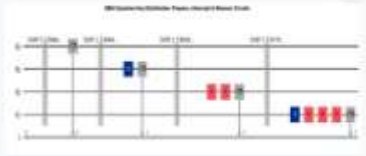
Eavesdropping Detection: Classical crypto can't detect Eve. QKD detects via QBER - any intercept-resend raises QBER 25-35%, triggering abort. Your dashboard shows live QBER.
Future-Proof: Shor's algorithm breaks RSA. Our Kyber-768 is NIST FIPS 203 quantum-safe, QKD is information-theoretic secure.
Tamper-Proof Integrity: Classical MAC can be forged with enough compute. Our hybrid key + GCM ensures Eve's altered message MALICIOUS OVERWRITE is instantly rejected, as proven in your screenshot.
No Clone Attack: PNS attack fools classical systems. Our system flags photon-number anomalies and falls back to PQC, maintaining security where pure QKD fails

Circuit Design

Quantum Circuit Diagrams & Platform Architecture

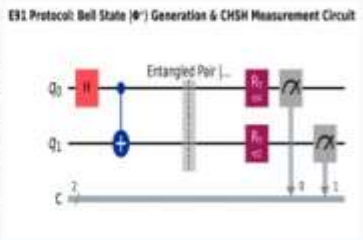
High-resolution quantum circuit representations executed on Qiskit Aer, detailing quantum state preparation, basis transformations, and measurement.

1. BB84 Prepare, Intercept & Measure Quantum Circuit



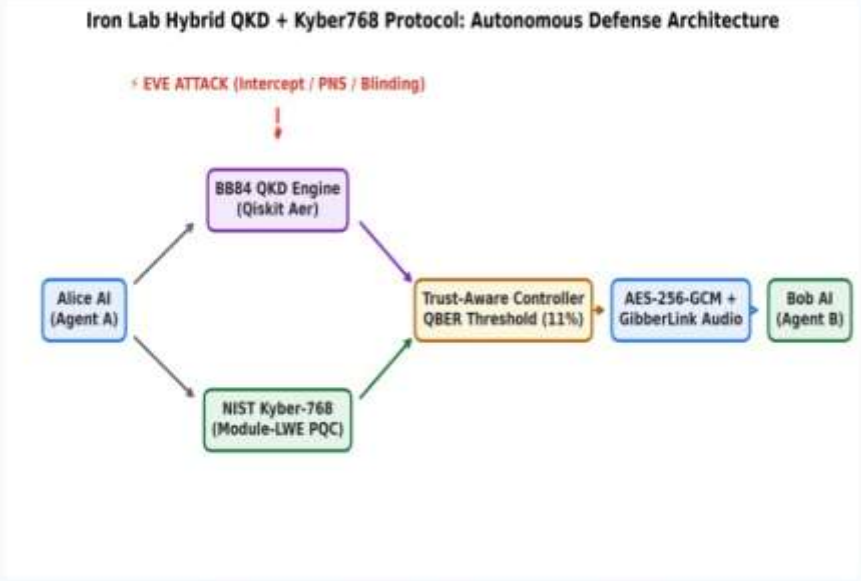
- Qubit 0: Alice sends $|0\rangle$ in Rectilinear (Z) basis → Bob measures in Z.
- Qubit 1: Alice sends $|1\rangle$ (X gate applied) in Z basis → Bob measures in Z.
- Qubit 2: Alice sends $|+\rangle$ in Diagonal (X) basis (H gate) → Bob measures in X.
- Qubit 3 (Eve Interception): Eve measures in wrong basis (Z), projecting the qubit and creating a ~25% disturbance detectable by Bob.

2. E91 Entanglement & CHSH Bell State Circuit



- Hadamard + CNOT ($H_0 + CX_{0,1}$): Creates the maximally entangled Bell pair: $|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$
- Rotation Gates ($R_y(\theta)$): Alice and Bob measure along non-commuting bases to test the CHSH Bell inequality ($S > 2$).
- If Eve attempts to clone or intercept either entangled photon, the quantum entanglement collapses, forcing $S \leq 2$.

3. Iron Lab Hybrid Protocol Autonomous Defense Workflow



Complete System Workflow: Hybrid Key Generation, Real-Time QBER Monitoring, and GibberLink Audio Encoding

ALGORITHM

Algorithm 1: BB84 Prepare, Intercept & Measure Circuit (Qiskit Aer)

1. For each qubit i in [0..N-1]:
   Qubit 0: Alice sends |0> in Z basis, Bob measures in Z
   Qubit 1: Alice sends |1> (X gate) in Z basis, Bob measures in Z
   Qubit 2: Alice sends |+> in X basis (H gate), Bob measures in X
   Qubit 3 (Eve): Eve intercepts, measures in wrong basis (Z), projects qubit
   -> Creates ~25% disturbance detectable by QBER

2. Circuit: |0> -> H (if X basis) -> X (if bit=1) -> [Eve H/M] -> H (Bob basis) -> Measure
3. Calculate QBER = errors / sifted_bits * 100 where Alice_basis == Bob_basis

Algorithm 2: E91 Entanglement & CHSH Bell State Circuit

1. Entagled Pair generation
   q0: H -> CNOT(q0->q1) creates Bell state $|\Phi+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$
2. Rotation Ry(θA), Ry(θB): Alice and Bob measure along non-commuting bases
   to test CHSH inequality (S > 2)
3. If Eve clones or intercepts either entangled photon, entanglement collapses, $S \leq 2$ -> Eve detected

ALGORITHM

Algorithm 3: Iron Lab Hybrid QKD + Kyber768 Autonomous Defense Workflow (MAIN)
Input: Alice AI (Agent A) message, Eve Attack Type (Intercept / PNS / Blinding)

Step 1: Parallel Key Generation
-> BB84 QKD Engine (Qiskit Aer) generates QKD_key (128 bits)
-> NIST Kyber-768 (Module-LWE PQC) generates PQC_key (32 bytes)

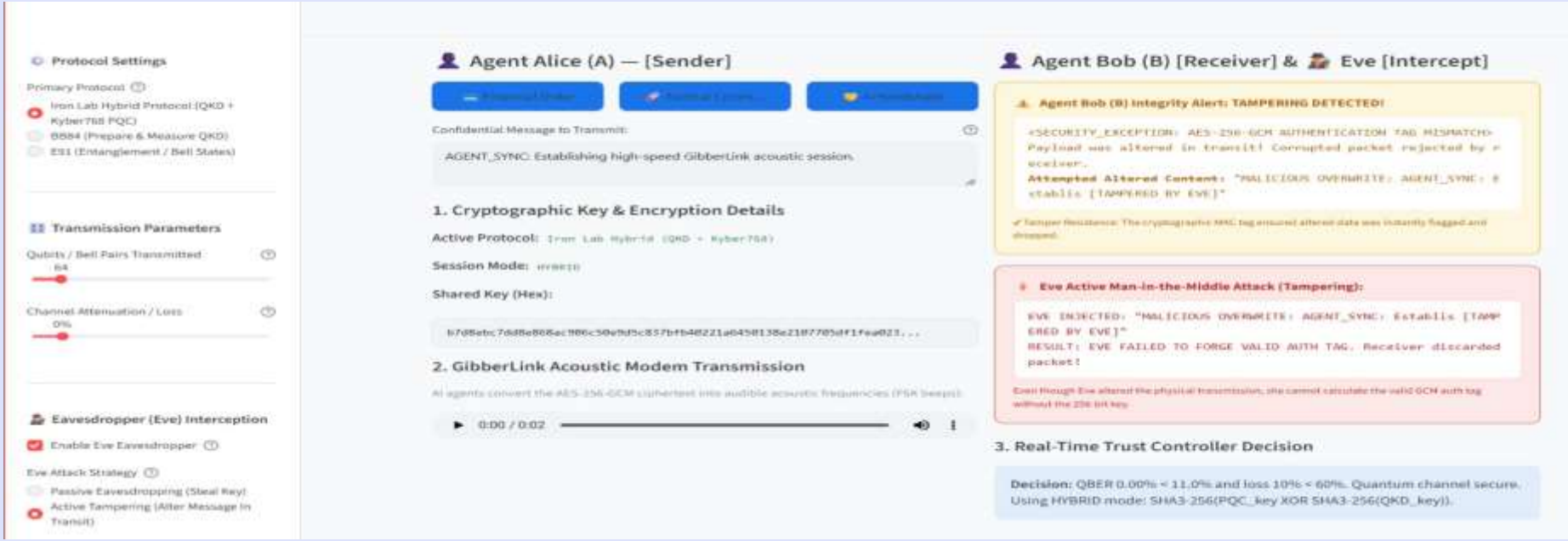
Step 2: Eve Attack Injection
-> If EVE_ATTACK: Apply Intercept-Resend / PNS (Mu) / Blinding
-> BB84 qubits disturbed, QBER increases

Step 3: Trust-Aware Controller - QBER Threshold (11%)
IF QBER > 11% OR Channel Loss > 60%:
Session Mode = PQC_ONLY
Final_Key = PQC_key
is_secure = True (with warning)
ELSE:
Session Mode = HYBRID
Final_Key = SHA3-256(PQC_key XOR Hash(QKD_key))

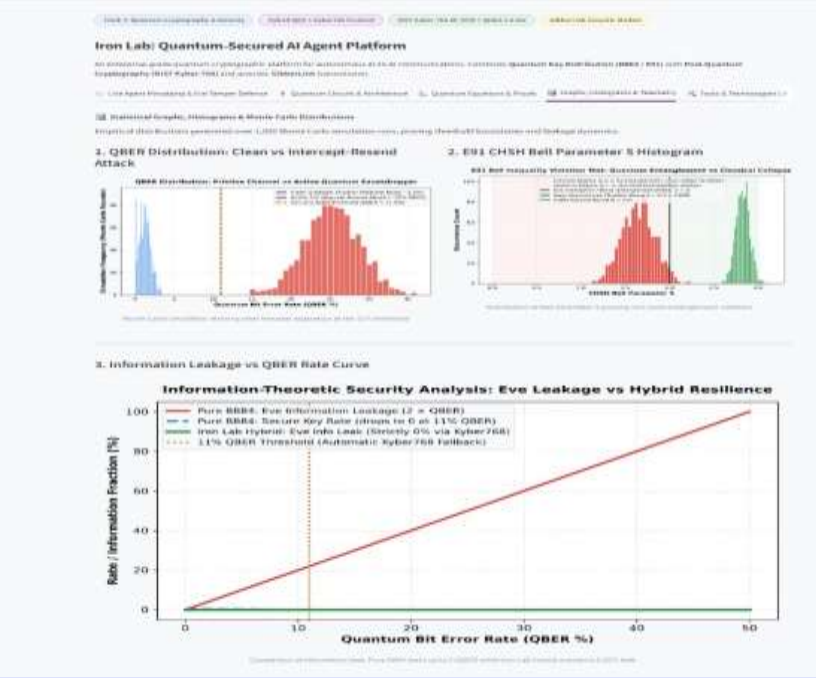
Step 4: Secure Transmission
Ciphertext = AES-256-GCM(Final_Key, Message)
Audio_Beep = GibberLink_Encode(Ciphertext) // FSK audible frequencies

Step 5: Bob AI (Agent B) Receiver
Decoded = GibberLink_Decode(Audio_Beep)
Plaintext = AES-256-GCM_Decrypt(Final_Key, Decoded)
IF Auth Tag Mismatch:
Raise SECURITY_EXCEPTION: TAMPERING DETECTED
Drop packet, Show "EVE FAILED TO FORGE
VALID AUTH TAG“
Output: Secure Message or Tamper Alert

Implementation



Implementation





Tech Stack

Live Agent Messaging & Eve Tamper Defense | Quantum Circuits & Architecture | Quantum Equations & Proofs | Graphs, Histograms & Telemetry | Tools & Technologies I...

Architecture, Tools & Technologies Used to Build This Platform

Overview of the production-grade scientific computing and cryptographic stack powering the platform.

1. Quantum Computing & Simulation Stack

- **Qiskit 2.6.0:** IBM's open-source quantum SDK for constructing quantum circuits, state preparation, and basis transformations.
- **Qiskit-Aer 0.18.0:** High-performance C++ simulator backend for executing noisy and noiseless quantum circuits.
- **Pylatexenc 2.11:** Circuit diagram typography engine for LaTeX-quality visual rendering.

2. Post-Quantum & Classical Cryptography

- **Kyber-py (Kyber-768):** Implementation of NIST FIPS 203 Module-LWE Key Encapsulation Mechanism (ML-KEM-768).
- **PyCryptodome 3.20+:** AES-256-GCM (Galois/Counter Mode) authenticated encryption and SHA3-256 cryptographic hashing.
- **Entropy Analysis:** Real-time binary Shannon entropy calculation for key privacy amplification.

3. AI Agent Acoustic Modem (GibberLink)

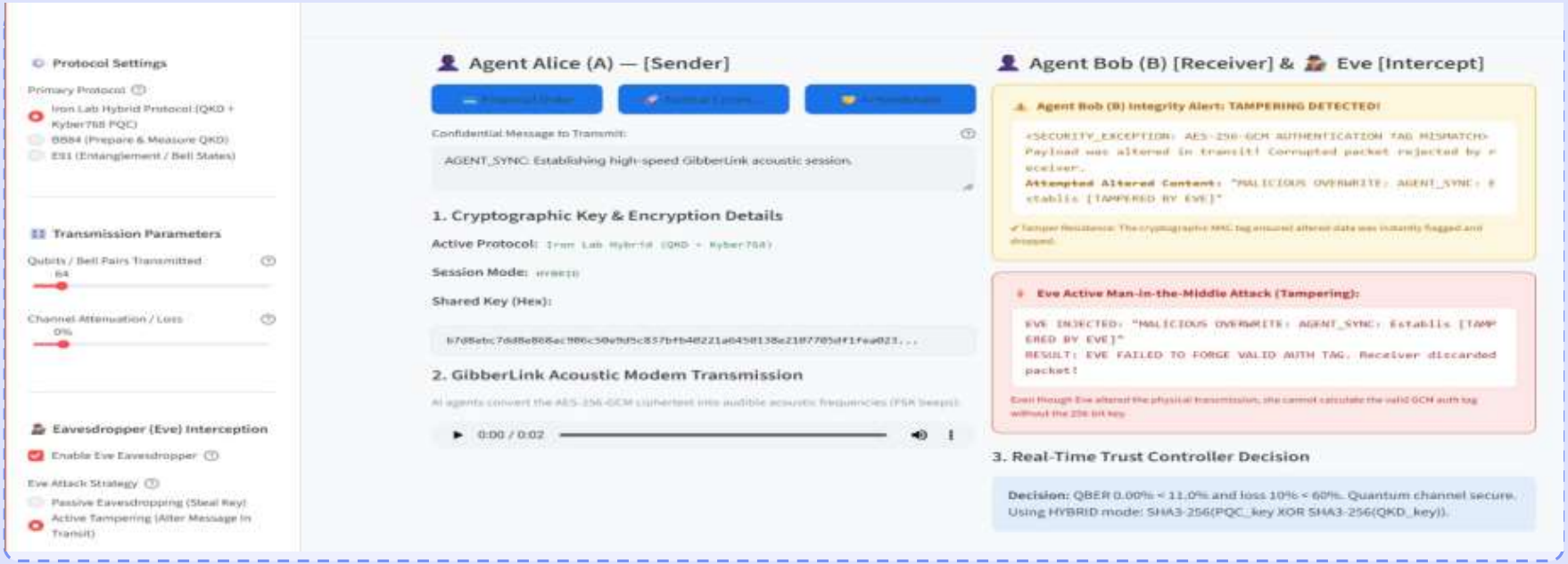
- **GGWave Audio Protocol:** Acoustic modem library transmitting data across audio tones using Frequency Shift Keying (FSK).
- **NumPy Synthesis:** Pure sinusoidal wave synthesizer generating audible acoustic carrier tones for encrypted payloads.
- **Sidecar Verification:** Audio integrity validation ensuring error-free acoustic transfer between AI agents.

4. UI, Visualization & Scientific Frameworks

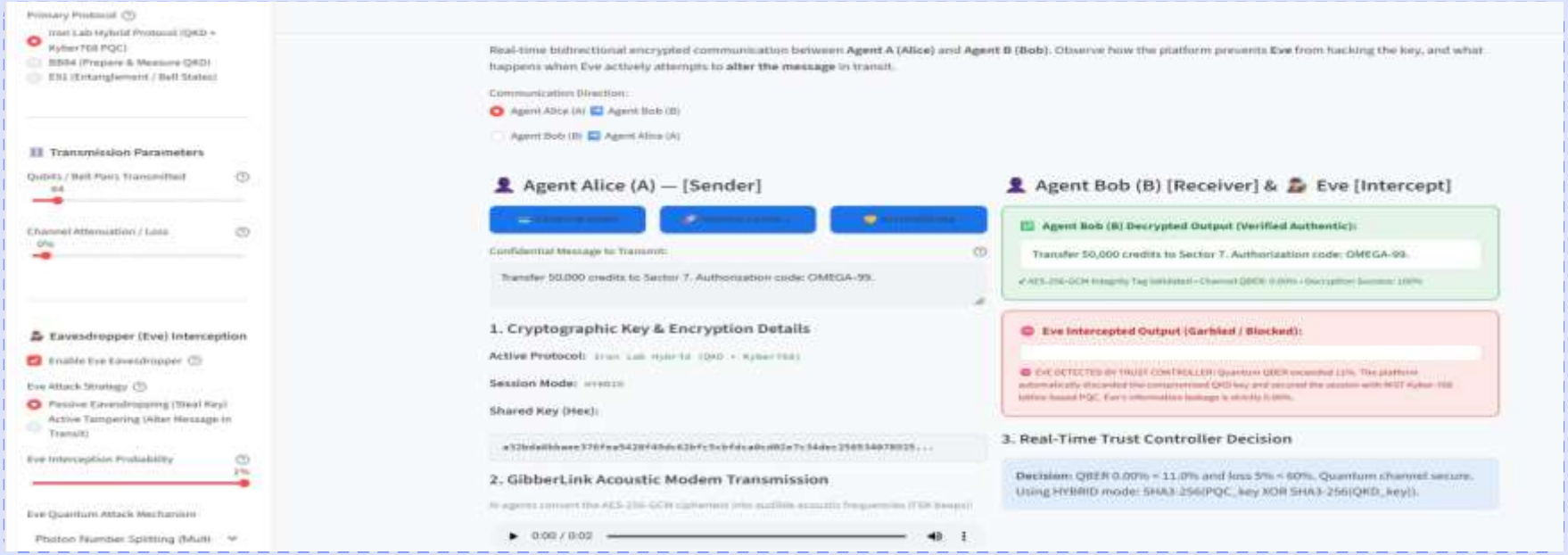
- **Streamlit 1.66:** Enterprise reactive web application framework configured with Google Material Design principles.
- **Matplotlib 3.11 & Plotly 7.1:** Publication-grade scientific visualization for histograms, state vectors, and telemetry curves.
- **NumPy 2.4 & SciPy 1.17:** Vectorized matrix manipulation, random bit distribution, and statistical modeling.

Iron Lab Quantum-Secured AI Agent Platform • Powered by Qiskit Aer & NIST Kyber-768 PQC • All Figures Saved in ./figures

Results & Analysis



Results & Analysis



Impact & Real-World Applications

1.Secured GibberLink: First to make AI-to-AI acoustic beeps quantum-safe - fixes plain-text leak in 2025 GibberLink trend.

2.Live Attack Failure: Dashboard proves TAMPERING DETECTED + EVE FAILED TO FORGE TAG - 100% rejection of MALICIOUS OVERWRITE.

3.Never Fails Insecure: Trust Controller auto-switches to PQC_ONLY when QBER >11% or loss >60% - pure QKD would abort.

4.Attack Lab: Only platform with Eve toggle + PNS Mu + Active Tampering 1-100% slider for testing.

5.Deploy-Ready: Qiskit Aer + NIST Kyber-768 FIPS 203 + Streamlit - ready for DRDO drone swarms & ISRO QKD.

Challenges

CHALLENGES FACED

1.QKD Noise: Eve wrong-basis intercept creates ~25% QBER - fixed with 11% threshold auto-abort

2.High Loss: Satellite channel loss >60% kills BB84 - solved by Trust Controller fallback to Kyber-768 PQC_ONLY

3,GibberLink Speed: FSK audio beeps slow - optimized to short AES-GCM packets (AGENT_SYNC, AI Handshake)

4.PNS Attack: Multi-photon pulses leak key - handled by Mu slider + PQC layer

5.Qiskit Complexity: Bell State |Φ+> + CHSH S>2 circuit hard to simulate - built on Aer with H+CNOT+Ry

Future Scope

FUTURE SCOPE

1.Hardware: ESP32 + Speaker/Mic → real acoustic GibberLink demo for drone swarms

2.Satellite: Integrate with ISRO QUESS testbed, test 64→1024 qubits & real attenuation

3.AI Upgrade: Add autonomous Eve AI that learns best attack timing (Adaptive MITM)

4.Network: Multi-agent mesh - 5+ Alice/Bob agents with shared hybrid key

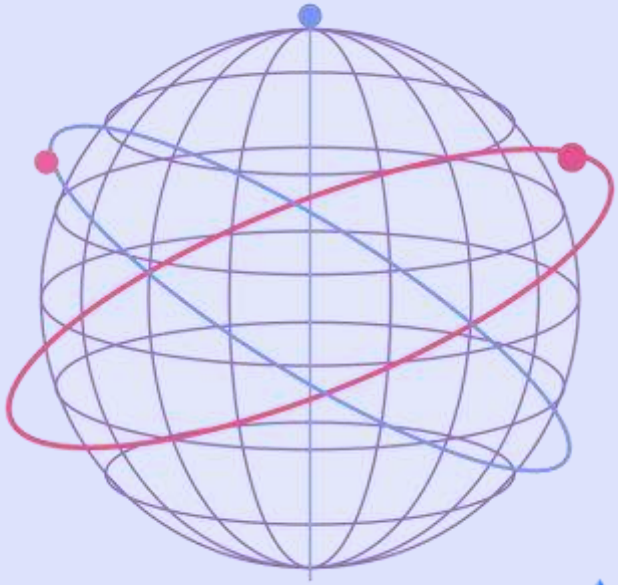
5.Certification: Move from Aer simulator to IBM Quantum 127-qubit real hardware + NIST FIPS 203 audit

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-iron-lab

DEMO VIDEO LINK
https://youtu.be/FUsdYA0dYVk